

Muhamad Nasir Lukman

GEOSPATIAL DATA SCIENCE AND REMOTE SENSING ANALYST

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Summary

Geospatial data scientist with five years of technical experience in the natural resource industry, specializing in integrating earth sciences with spatial data science to drive informed business decisions. Over one year in a managerial role with project management responsibilities, alongside two years of research in advanced remote sensing methodologies. Strongly interested in combining natural sciences, technology, data-driven approaches, and interdisciplinary collaboration for sustainable development.

Experience

Geospatial and Remote Sensing Specialist

5/2025 – present

Forest Carbon

- Building automated rapid carbon project assessment tools.
- Building national level AI model for annual landcover classification.
- Conducting more detailed modelling and analysis at project level for project development and monitoring related with geospatial and remote sensing data.

Geospatial Data Science and Remote Sensing Consultant

11/2024 – 12/2025

United Nations Office on Drugs and Crime (UNODC)

- Develop a proof of concept for detecting and classifying mining areas with remote sensing data, utilizing deep learning and machine learning techniques.
- Establish partnerships with relevant Indonesian agencies to identify innovation opportunities and data collection requirements.
- Combine remote sensing with other geospatial data layers to analyze compliance, environmental conditions, and the socioeconomic context of mining areas in Indonesia. This analysis will provide statistics on the dynamics of illegal mining sites, including their locations and their impact on environment and local communities, in particular related to critical raw material.
- Planning and delivering workshops with key government officials related to the state-of-the-art solution of geospatial and remote sensing analysis for mining monitoring.

Teaching Assistant

03/2023 – 03/2024

ITC University of Twente

- Conducted extra classes for three courses: Geodata Acquisition and Management, GIS & Remote Sensing for Geospatial Solutions, and Geospatial Analysis and Interpretation.

Research Geologist

10/2021 – 6/2022

Faculty of Engineering, Universitas Gadjah Mada Research Center (LKFT)

- Performed fieldwork, laboratory work, and data analysis for detailed regional geological, stratigraphic and tectonic investigations of the South Sumatra and East Borneo Basins.

Project Geologist

8/2018 – 6/2021

TIMAH

- Primary tin deposit exploration which includes planning, executing, and evaluating soil sampling, field mapping, anaconda mapping, trenching, and drilling campaign.
- Mapped informal mining activities within company concessions using satellite imagery, modeled recovery efficiency, and assessed reserve loss due to illegal mining activity.
- Developed a statistical model incorporating geological uncertainties and operational variables to enhance reserve estimations and mine planning for offshore placer deposits using Monte Carlo simulations to improve financial model transparency and reliability.
- Automate database format transformations and created a machine-learning model to predict missing values in the exploration database. Enhanced visualization schemes for drill logs and soil sampling datasets.
- Enhanced the geodatabase and led a redrilling project to improve data quality for decision-making

Research Assistant

1/2015 – 1/2016

Geothermal Research Center of Universitas Gadjah Mada

- Conducted routine laboratory analysis of geothermal wells samples, prepared workshops, coordinated fieldwork, and prepared project reports and presentation drafts.

Education

University of Twente, MSc Geoinformation and Earth Observation 9/2022 – 7/2024

- Research: Bayesian spectral unmixing for sub-pixel image classification (mineral quantification) of core sample from hyperspectral imagery. Graduated *cum laude*

Universitas Gadjah Mada, BEng Geological Engineering 7/2011 – 5/2016

- Research: Geological structure analysis with IFSAR lineament extraction, field measurement, and earthquake focal mechanism data. Graduated with a *very good* remark

Honors & Awards

- Change Leaders and Management Honors Student – University of Twente
- Selected Project for TINS HACKATHON 2019 – TIMAH
- Presenter in Geology of Southeast Asia XIV Congress 2016 – Universitas Gadjah Mada
- Selected Student Research Grant Award – Universitas Gadjah Mada

Skills

- **Programming:** Experienced in Python, R, and JavaScript for data analysis.
- **Data Science:** Competent in machine learning, deep learning, physical modeling, and Bayesian modeling for data analysis.
- **Domain Knowledge:** Strong background in earth systems and extensive experience in the mining industry, including knowledge of regulations, operational practices, and global commodity trends. Strong knowledge in Verra Voluntary Carbon Standard for Nature Based Solutions.